

Engineering Masterclass 002

Career Intelligence Copilot — FR-017

Agent Evaluation & Observability

Derive-Only Metrics, Reconstructability, and Anti-Theatre Observability

ENGINEERING LEARNING ACADEMY

LEAN EDITION

STATUS: COMPLETE / FROZEN / ACCEPTED

Executive Summary

FR-017 answers a deceptively simple question: once a multi-agent orchestration layer has written honest audit records, how do you **evaluate** those records without inventing an observability product?

The accepted answer is a **narrow derive-only evaluation substrate**. Existing `OrchestrationRun`, `Handoff`, and child `AgentRun` artefacts are the only evidence. Pure functions derive metrics, correlate parent/child execution, and score reconstructability against twelve fixed questions (R1-R12).

The Core Posture

- Low near-term product value
- High learning and interview value
- Nothing mutates the runtime
- Nothing becomes a second system of record
- Horizon 1B is **not** gated on this work

📄 **The engineering win:** Honesty under uncertainty — especially the distinction between *missing* metadata and *measured zero*, and the refusal to repair contradictory audits.

The Engineering Problem

After FR-016, Career Intelligence Copilot had orchestration audits: parent runs, typed handoffs, stop reasons, child agent runs, and resume/idempotency keys. What it lacked was a first-class way to **use** those audits as evaluation evidence.

1

Reconstruct Authority

Can I reconstruct the authority story of a run without reading source code?

2

Offline Comparison

Can I compare runs offline, deterministically, without a live LLM?

3

Missing vs. Zero

When tokens or cost are absent, is that "zero usage" or "unknown"?

4

Honest Gaps

If a child id does not join cleanly to a handoff, do we invent linkage or surface the gap?

📌 These are **evidence, semantics, and source-of-truth** problems — not "add Prometheus" or "build a Grafana board" problems.

Why Previous Approaches Were Insufficient

The Laundry-List Observability FR

Traces, checkpoints, dashboards, richer telemetry — most already owned by FR-008, FR-014, FR-015, FR-016. Building the catalogue would have duplicated frozen work and delayed Horizon 1B for no acquisition gain.

Instrument the Runtime

Adding events until every chart is easy fails the derive-only gate. If evaluation requires new runtime events, you're evaluating a redesigned system — not the current one.

Fork Child Metrics

BOPA already had `AgentRunMetrics` with careful missing-vs-zero semantics. Re-implementing inside FR-017 creates two competing definitions of the same facts.

Dashboards as Proof

A board plotting "latency = 0" for runs that never recorded latency converts absence into fake precision. Dashboards create the *feeling* of control — not reconstructability.

Block Horizon 1B

Coupling recruiter outreach to FR-017 inverts sequencing priorities. Evaluation substrate improves learning; it does not improve application throughput.

Ownership Matrix — What FR-017 Does Not Own

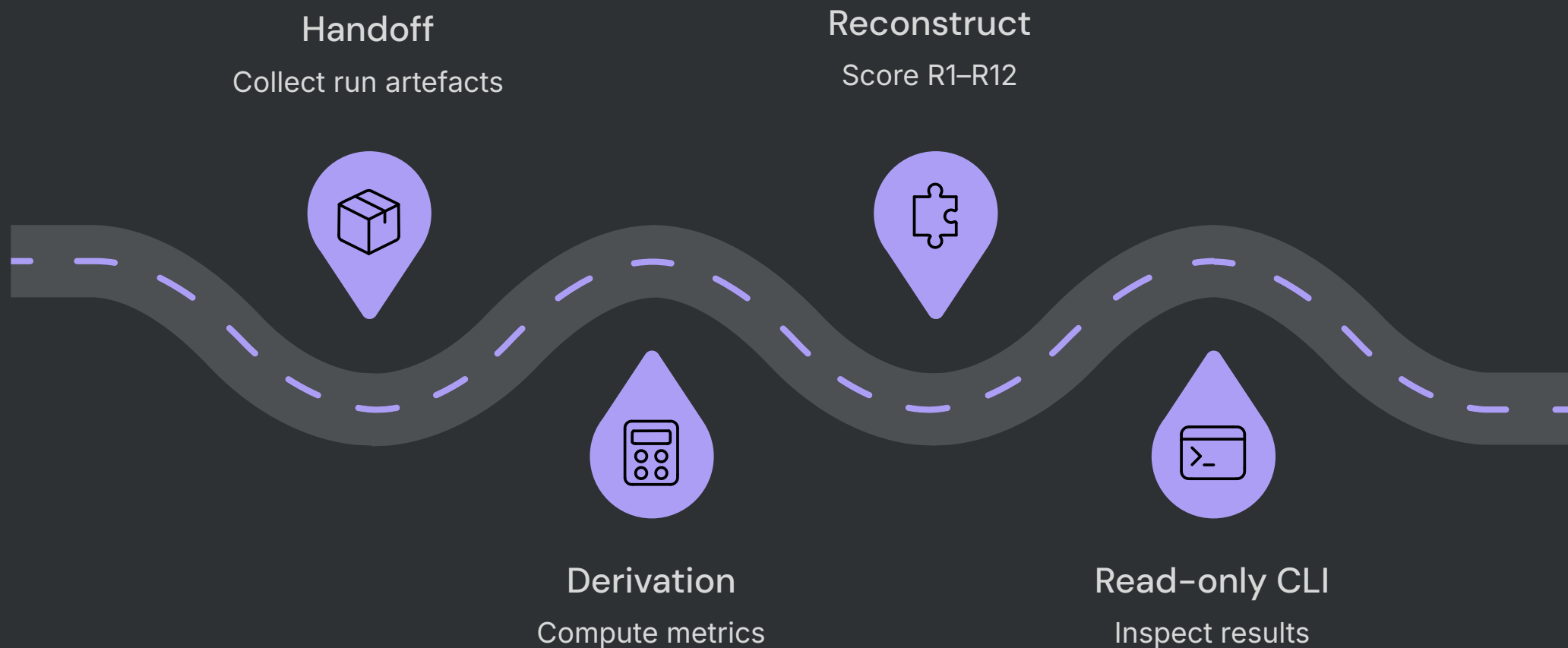
FR-017's scope is deliberately narrow. Understanding what it *does not* own is as important as understanding what it does.

Concern	Owner	FR-017 Role
Workflow traces	FR-008	None
Truth engine	FR-014	Cite status only
BOPA / AgentRun metrics	FR-015	Reuse
Orchestration audits	FR-016	Source records
Derive, correlate, evaluate, present	FR-017	Evaluation only

📄 Crossing ownership boundaries creates twin definitions and silent drift. FR-017 evaluates — it does not become a system of record.

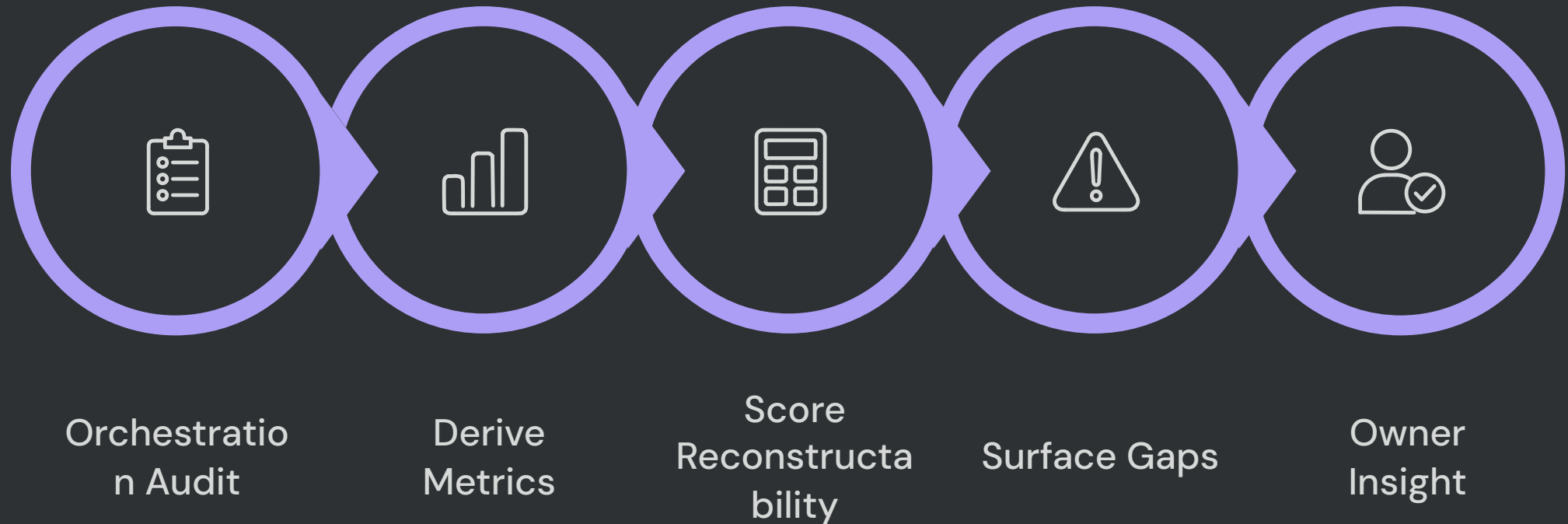
The Chosen Architecture

The architecture is deliberately boring. That is the point.



No store writes. No supervisor start or resume. No specialist execution. No telemetry store. No new audit event kinds. No live tracing. No alerts. No pipeline mutation.

Runtime Example — Evaluation Flow



What the module extracts

- Per-run metrics: identity, goal, status, stop reason, limits
- Handoff metrics: allow/deny, policy fields, lifecycle
- Optional child roll-ups: provider, tokens, cost, latency (None when absent)
- Parent/child correlation including orphans and join gaps

CLI surface

- `cic agent orchestrate metrics` — one run
- `cic agent orchestrate metrics-corpus` — offline suite summary

Presentation is subordinate to derivation. The CLI is an interview and debug surface, not a product dashboard.

R1–R12 Reconstructability Questions

Twelve fixed questions an owner must be able to answer from artefacts alone. **PASS** = supported by evidence. **FAIL** = names the missing or contradictory evidence. The evaluator never invents precision and never "heals" bad audits.

ID	Story Beat	What Must Be Evidenced
R1	Owner goal	Goal recorded and readable from artefact
R2	Observed state	Observation snapshot present
R3	Specialist selection	Selection decision traceable
R4	DelegationPolicy consistency	Policy fields consistent across handoff
R5	Authority boundary	OBS / BOPA boundary respected
R6	Handoff lifecycle	Allow/deny and lifecycle fields present
R7	Child output references	Child output joinable to parent
R8	Stop reason	Stop reason recorded and typed
R9	Owner next action	Next action derivable from artefact
R10	Limits	Limit fields present and evaluable
R11	Parent/child walk	Full parent→child chain walkable; orphans surfaced
R12	Resume / idempotency	Resume key present and consistent

Key Engineering Principles



Derive Before Instrument

Prove what existing records support. Instrument only when reconstructability fails for a reason that cannot be fixed by better derivation.



Missing $\neq$ Zero

`None` = field never recorded. `0` = measured empty. Coercing absence into zero is a semantic bug that infects every aggregate downstream.



Preserve Source-of-Truth Boundaries

FR-017 evaluates. FR-015 owns child metrics schemas. FR-016 owns orchestration audit shape. Crossing boundaries creates twin definitions and silent drift.



Observability $\neq$ Dashboards

Observability means: can a competent engineer reconstruct what happened and what was unknown? Charts are one possible presentation — not the definition.



Detect, Don't Repair

Orphan child ids and contradictory linkage produce R-signal failures. Evaluation surfaces them. Repair belongs to runtime correctness work.



Read-Only Evaluation

An evaluation CLI that can start supervisors or mutate pipelines is a control plane with a misleading name. FR-017 paths load only.

Validation Summary

15/15

Corpus PASS

Deterministic cases; repeatability confirmed

GO

Recommendation

ACCEPT AND FREEZE — remain narrow
derive-only

R12

Questions Scored

Full reconstructability story from goal to
resume/idempotency

Corpus shapes covered

- Successful orchestration and blocked delegation
- BOPA child and OBS brief, including `prepare_then_brief`
- Missing optional metadata vs. measured zero
- Orphan child, missing child join, stale handoff
- Resumed / idempotent run; loop stops; contradictory audit

Constraints confirmed

- No DOS/BOPA/OBS contract changes
- No dashboards or telemetry SoT
- Horizon 1B unblocked
- Simpler daily prep path preferred
- Byte-identical proof: metrics commands do not mutate stored JSON

✓ Expected failures are part of the design: orphan cases fail R11; contradictory audits fail R7 and R11. A suite that only passes happy paths would be reconstructability theatre.

Trade-offs

✓ Accepted

Choice	Why Worth It
Derive-only over existing audits	Tests the real system, not a redesigned one
R1-R12 as first-class acceptance	Forces evidence, not storytelling
Reuse FR-015 child metrics	One definition of tokens/cost/provider
Thin CLI instead of UI	Interviewable and owner-operable without product weight
Low commercial claim	Avoids observability theatre and false ROI

⌚ Deferred / Out of Scope

- Token/cost completeness for every offline fixture
- CLI polish and richer fault-injection productisation
- Dashboards, alerts, live tracing platforms
- Telemetry backends, monitoring SaaS
- Pipeline mutation, submission behaviour
- Horizon 1B integration

📄 Richer time-travel replay or broader observability — **only** when derive-only evaluation fails to answer owner-critical questions at scale. Desire for dashboards is not justification.

Honest Product Assessment

This honesty is itself an engineering artefact. Overclaiming value is a form of technical debt in product communication.

Improves daily preparation? No material improvement	Materially improves Horizon 1B? No	Useful operational / audit insight? Yes
Justifies dashboards now? No	Should remain narrow? Yes	

Why Employers Care

Employers need engineers who can evaluate complex systems **without inventing parallel platforms**. FR-017 demonstrates transferable production-grade judgment:

- **Derive evidence from existing records**
Don't redesign the system to make it measurable.
- **Refuse false precision when data is missing**
Honest semantics over pretty aggregates.
- **Detect parent/child inconsistency without silently fixing it**
Surface contradictions; repair belongs to runtime correctness work.
- **Keep evaluation read-only**
Diagnosis must not become accidental mutation.



Interview Preparation — Q&A

Use these as spoken answers, not memorised paragraphs.

What problem did FR-017 solve?

How to evaluate multi-agent orchestration audits honestly — reconstructability, missing-data semantics, parent/child correlation — without building an observability platform or changing the runtime.

Why not add dashboards first?

Dashboards present numbers; they do not create trustworthy semantics. If missing latency is plotted as zero, the board lies. Prove derive-only reconstructability first.

What does derive-only mean in practice?

Pure functions over existing `OrchestrationRun` / `Handoff` / child `AgentRun` records. No new events, no telemetry store, no DOS/BOPA/OBS behaviour changes.

Explain missing versus zero.

`None` = optional field never recorded. $\emptyset$ = recorded and measured empty. Aggregates must not coerce absence into zero or every cost roll-up becomes fiction.

Give an example of an expected FAIL.

An orphaned child reference fails R11. A contradictory completed run that cannot join child output fails R7 and R11. The corpus treats those as GO when detected correctly.

When would you build richer observability?

When owner-critical questions cannot be answered from derived audits at scale — with concrete failing evidence, not aesthetic preference for dashboards.

More Interview Q&A

What are R1–R12?

Twelve reconstructability questions from goal through resume/idempotency. PASS requires evidence; FAIL names the gap. Contradictions are surfaced, not repaired.

Why reuse FR–O15 metrics instead of redefining them?

Child token/cost/provider semantics already existed. Forking them would create two truths. Evaluation should consume authoritative metrics, not re-author them.

Is this the daily operator path?

No. Ordinary preparation remains the simpler bounded agent command. FR-017 is learning/substrate and audit evaluation.

Did this block the next product horizon?

Explicitly no. Application-loop usability unblocks Horizon 1B, not completion of every learning FR.

What is the main risk this design avoids?

Observability theatre: instrumenting and charting until the system *looks* monitored while reconstructability and semantic honesty remain unproven.

Three Engineering Lessons to Remember



1. Derive Before You Instrument

Probe what existing records already support. New telemetry is a design change, not a default first step.



2. Missing Is Not Zero — Contradictions Are Not Repaired by Evaluation

Honest semantics beat pretty aggregates. An evaluation layer that heals bad audits is lying for convenience.



3. Observability Is Reconstructability Under Authority Boundaries — Not Dashboards

If you cannot walk goal → decision → handoff → child → stop → next action from artefacts, you do not have observability yet, no matter how many panels you ship.



The FR-017 Posture — What Senior Engineers Defend

Separate Evaluation from Runtime Mutation

Evaluation paths load only. They do not start supervisors or mutate pipelines.

Refuse Second Sources of Truth

FR-017 evaluates. FR-015 and FR-016 own their schemas. No twin definitions.

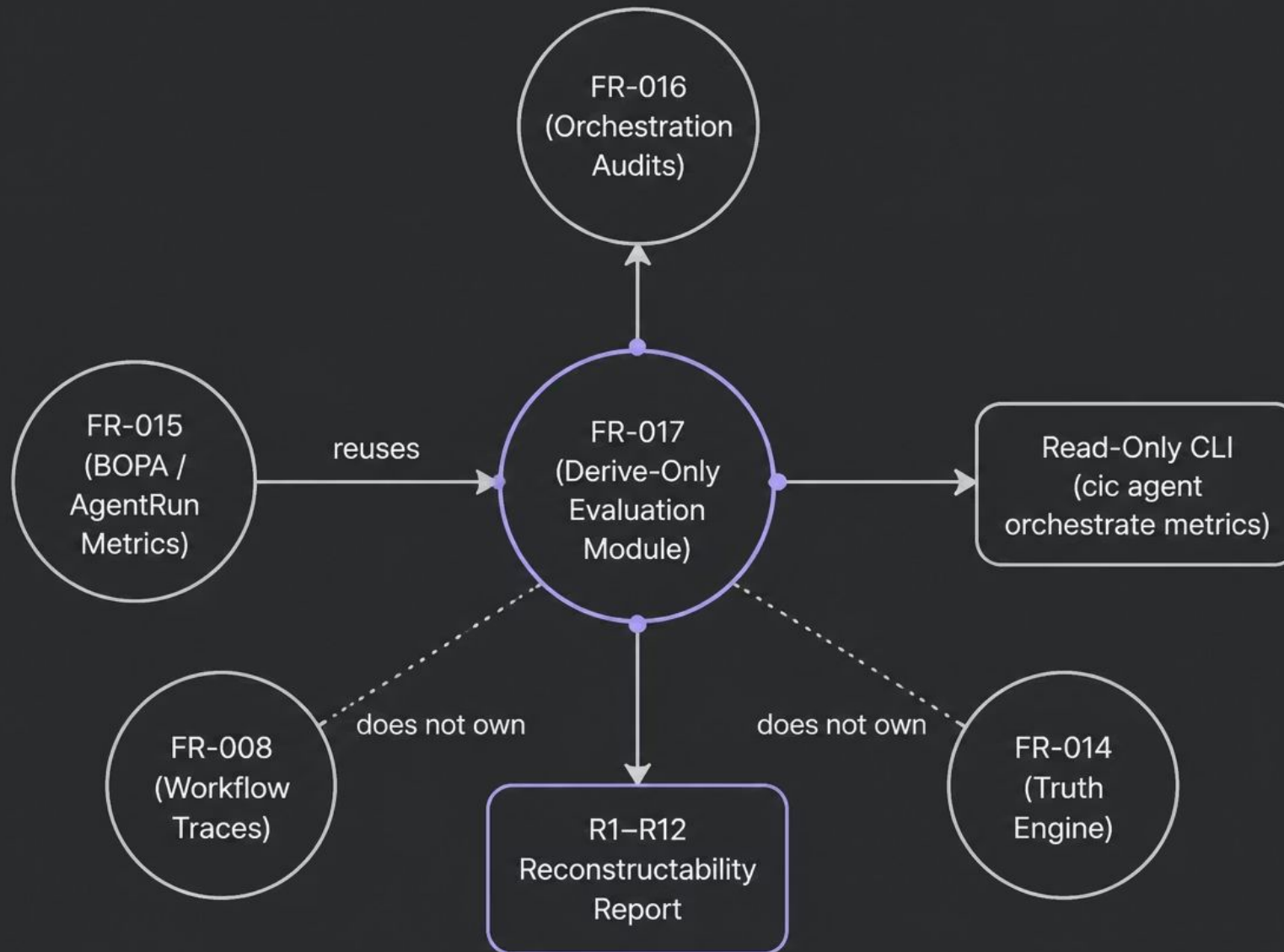
Keep Commercial Claims Proportional to Evidence

Low near-term product value is stated honestly. Overclaiming is technical debt in product communication.

Leave Adjacent Horizons Unblocked

Learning work not on the critical path must not gate acquisition work. Sequencing discipline is an engineering principle.

Architecture at a Glance



FR-017 sits at the centre of evaluation — consuming from FR-016, reusing from FR-015, and presenting through a thin read-only CLI. It touches adjacent FRs only to cite status, never to own their data.

The Interview-Ready Story

Not "we built observability" — but "we made orchestration audits evaluable: honestly, narrowly, and without theatre."

Chosen Architecture

Derive-only R1-R12, read-only CLI.

Validation

15/15 GO, deterministic, no instrumentation.

Engineering Problem

Audits existed but were not evaluable.

Engineering Posture

Honest, narrow, Horizon 1B unblocked.

FR-017 is a small system with a large teaching surface. It freezes a posture that senior engineers are asked to defend verbally in interviews: separate evaluation from runtime mutation, refuse second sources of truth, keep commercial claims proportional to evidence.

Memorable Closing Statement

Derive the truth you already have —

before you instrument a truth you wish you had.

FR-017

Agent Evaluation & Observability —
Complete / Frozen / Accepted

Engineering Learning Academy

Lean Edition Masterclass — canonical
acceptance report and ADR-009 remain
authoritative

Audience

Experienced software engineers
preparing for technical interviews on
multi-agent systems and observability